

The Mercury Cycle

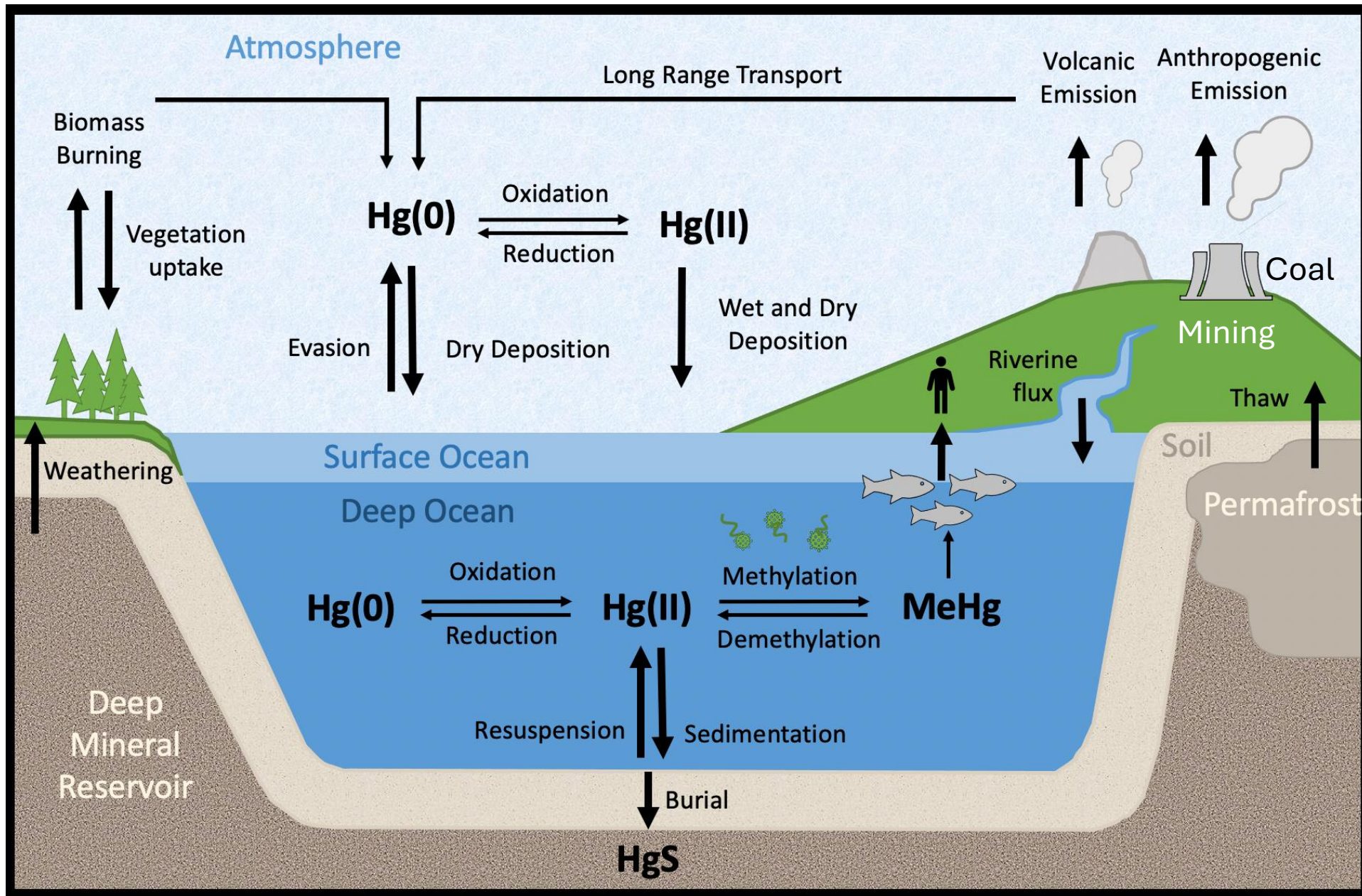
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EES301

Introduction

- Mercury (Hg) is found as a **trace metal in igneous rocks** and is locally concentrated in deposits of a red mineral - cinnabar (HgS).
- Volcanoes are a large natural source of Hg emissions to Earth's atmosphere, in the form of elemental mercury, **Hg⁰—a volatile metal**
- **Mercury contamination, particularly in the form of methylmercury**, can lead to severe neurological damage, developmental problems in humans.
- In ecosystems, it can **disrupt food webs, reduce biodiversity, and impair ecosystem services**.

Key Mercury forms

- **Elemental mercury (Hg^0):** predominant form in the atmosphere; volatile and relatively low reactive.
- **Inorganic mercury (Hg^{2+}):** more reactive and prevalent in aquatic systems and soils.
- **Organic mercury:** primarily methylmercury (MeHg), the most toxic form.



Mercury cycle

- Methylation is primarily a biotic process, whereas demethylation can happen through both biotic and abiotic processes.

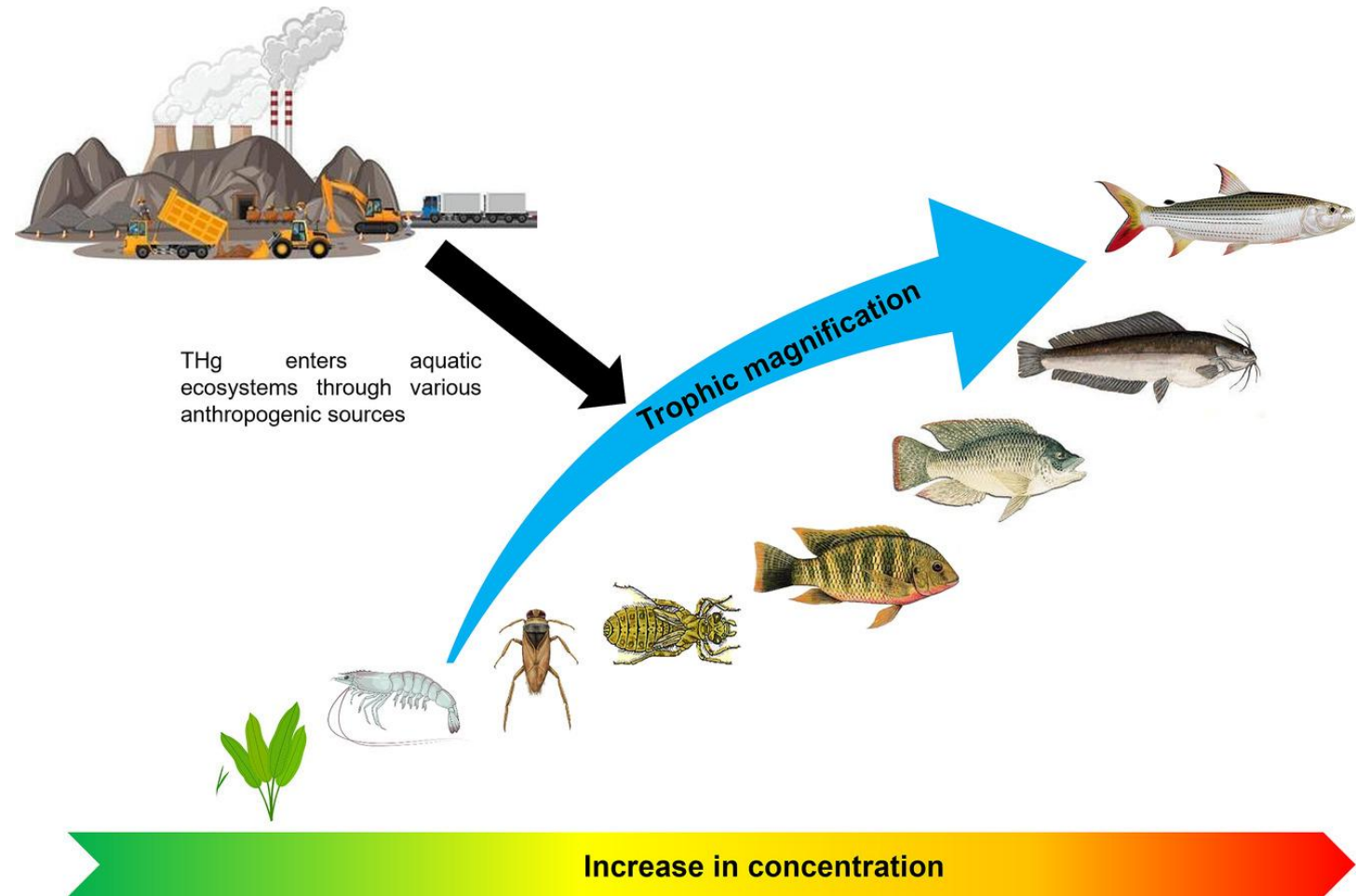
Key processes

- **Release/Emissions:** from volcanoes, mining, industrial processes, and waste incineration.
- **Transport:** atmospheric transport distributes gaseous Hg^0 globally. Water currents and river systems play a role in local and regional Hg transport.
- **Deposition:** onto land and water through precipitation and dry deposition.
- **Transformation:** In aquatic environments, Hg(II) can be transformed into methylmercury, a highly toxic organic form, through microbial processes.

- **Bioaccumulation and Biomagnification:**

Methylmercury accumulates in organisms, starting from phytoplankton and zooplankton and moving up the food chain.

- **Re-emission and Burial:** re-emission back into the atmosphere; a portion of mercury is buried in sediments - effectively removing it from the active cycle for geological timescales.



Rooyen et al. Science of The Total Environment
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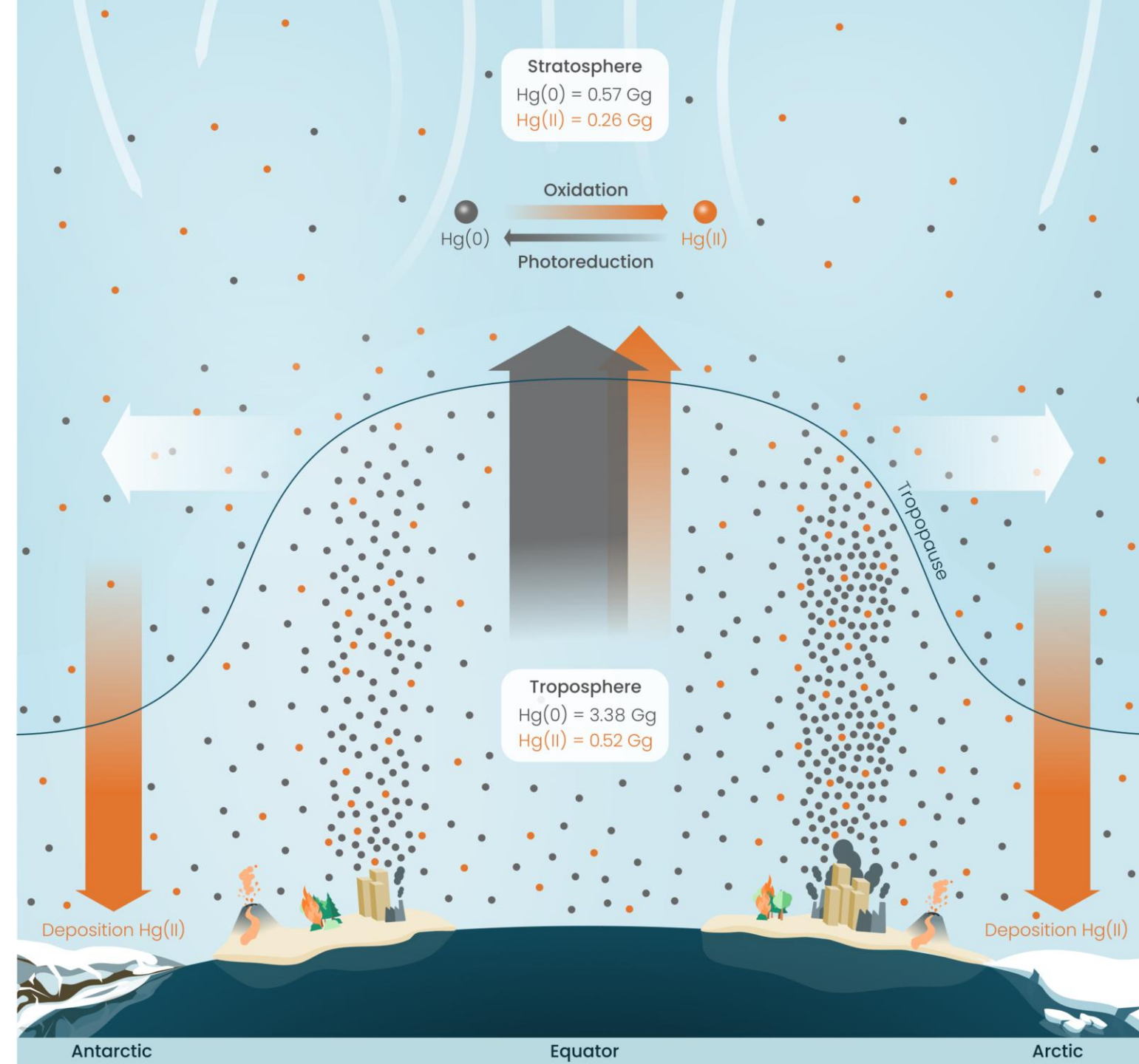
Atmospheric processes

- Elemental mercury has a long residence time – **long distance transportation** - meaning that Hg released in one location can have environmental consequences far away.
- The lifetime of Hg in the atmosphere ranges from **about one year for Hg⁰ to a few weeks for Hg²⁺**.
- Chemical processes like oxidation can **convert elemental mercury to reactive gaseous mercury (RGM), equivalent to Hg²⁺** which is more readily deposited.

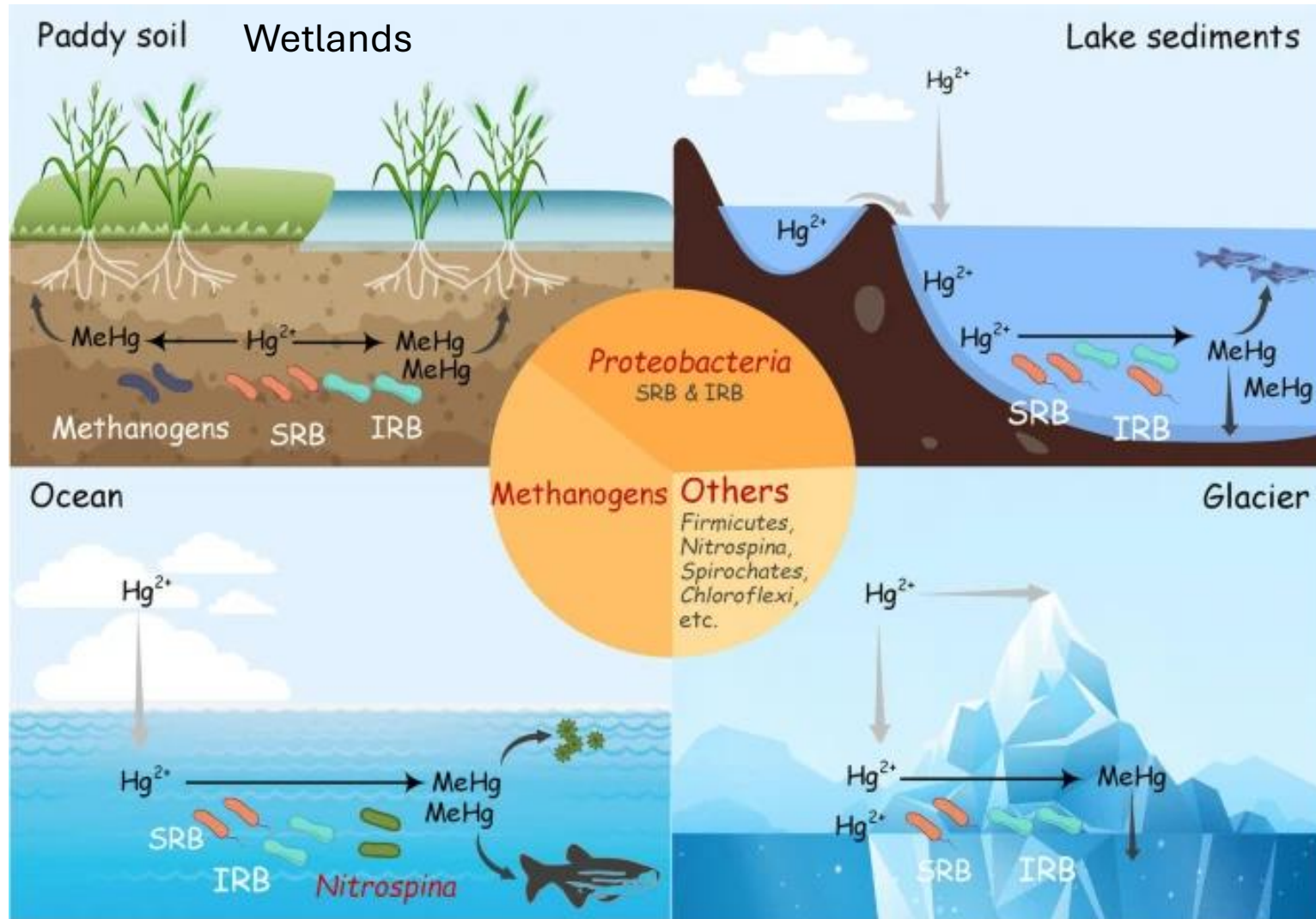
Conceptual representation of the exchange of mercury across the tropopause and surface deposition fluxes

- The transport of mercury to the stratosphere is predominantly as elemental $\text{Hg}(0)$ (**black dots**) through the tropics and midlatitudes (20° to 60°), while most of the stratospheric mercury subsides back into the troposphere at midlatitudes and the polar regions, with a larger fraction of oxidized mercury (**orange dots**).

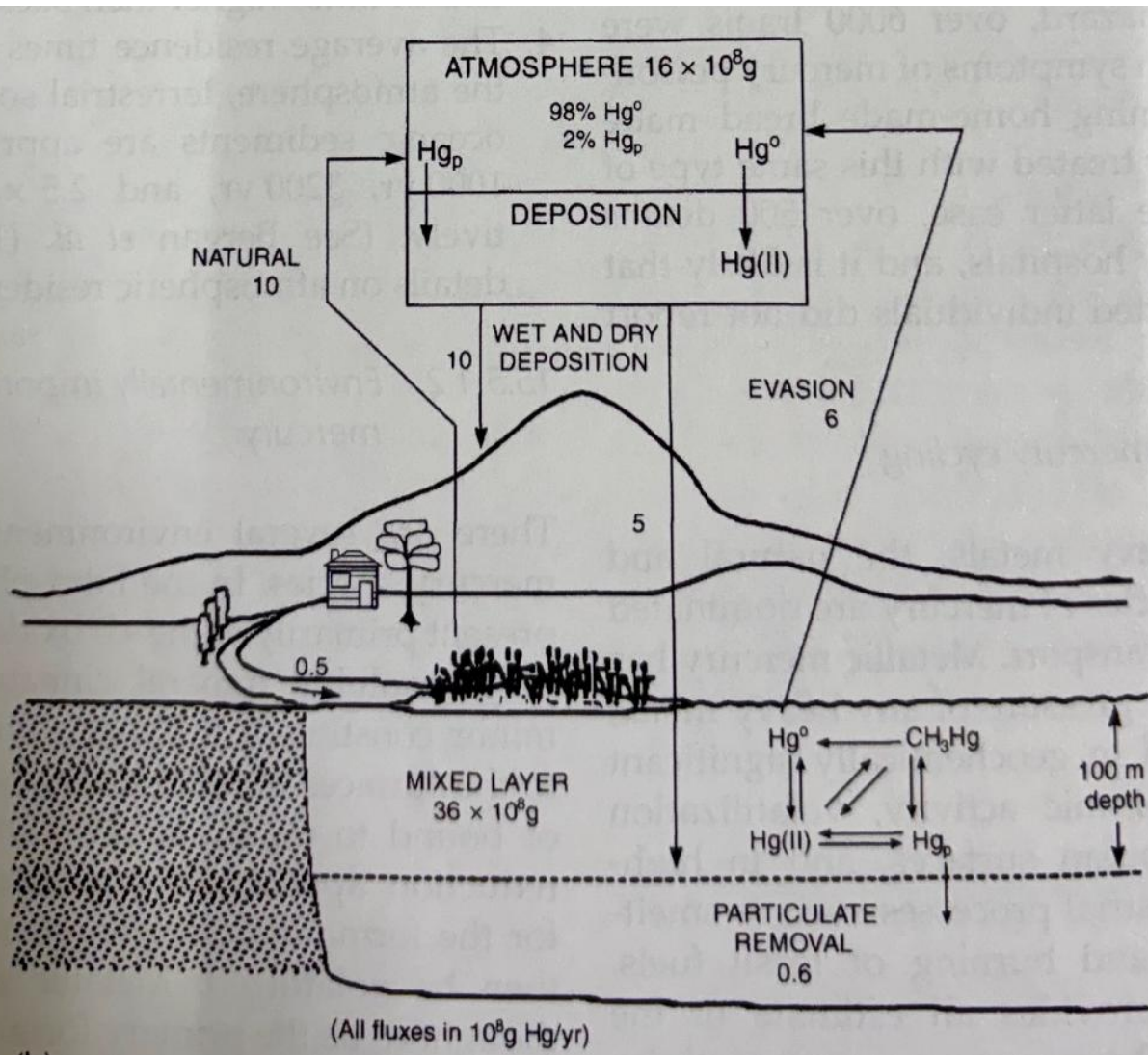
Science Advances, 2025, Vol. 11, No. 2
Role of the stratosphere in the global mercury cycle (Saiz-Lopez et al.)



Hg - Methylation



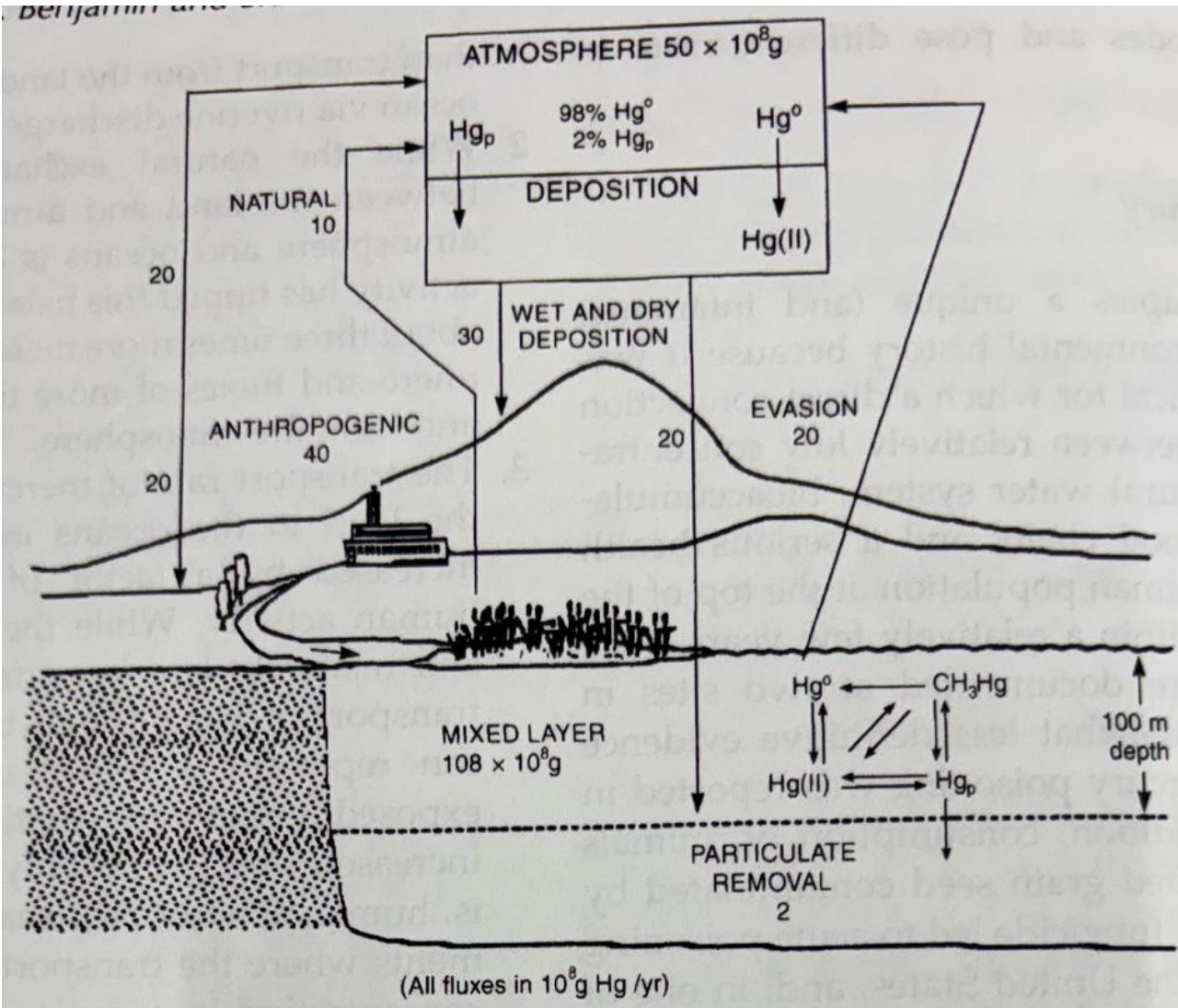
- Methylation is microbially-mediated process.
- IRB (Iron reducing bacteria) and SRB (Sulfate-reducing bacteria) are primarily involved in producing methyl-mercury in sediments - **cycles of S (+Fe) and Hg are linked by metabolism.**



Pre-industrial

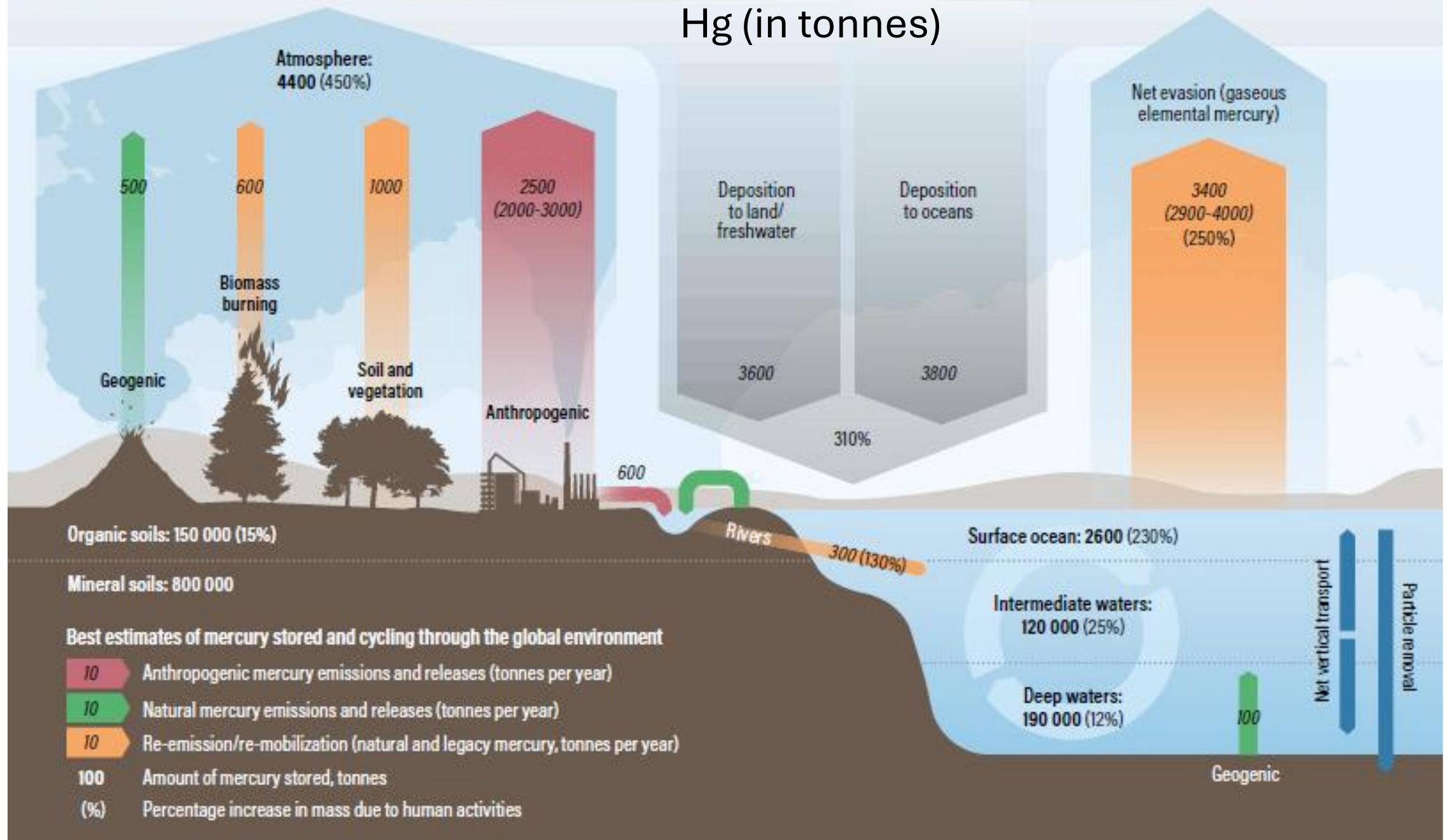
Global burdens and fluxes of mercury

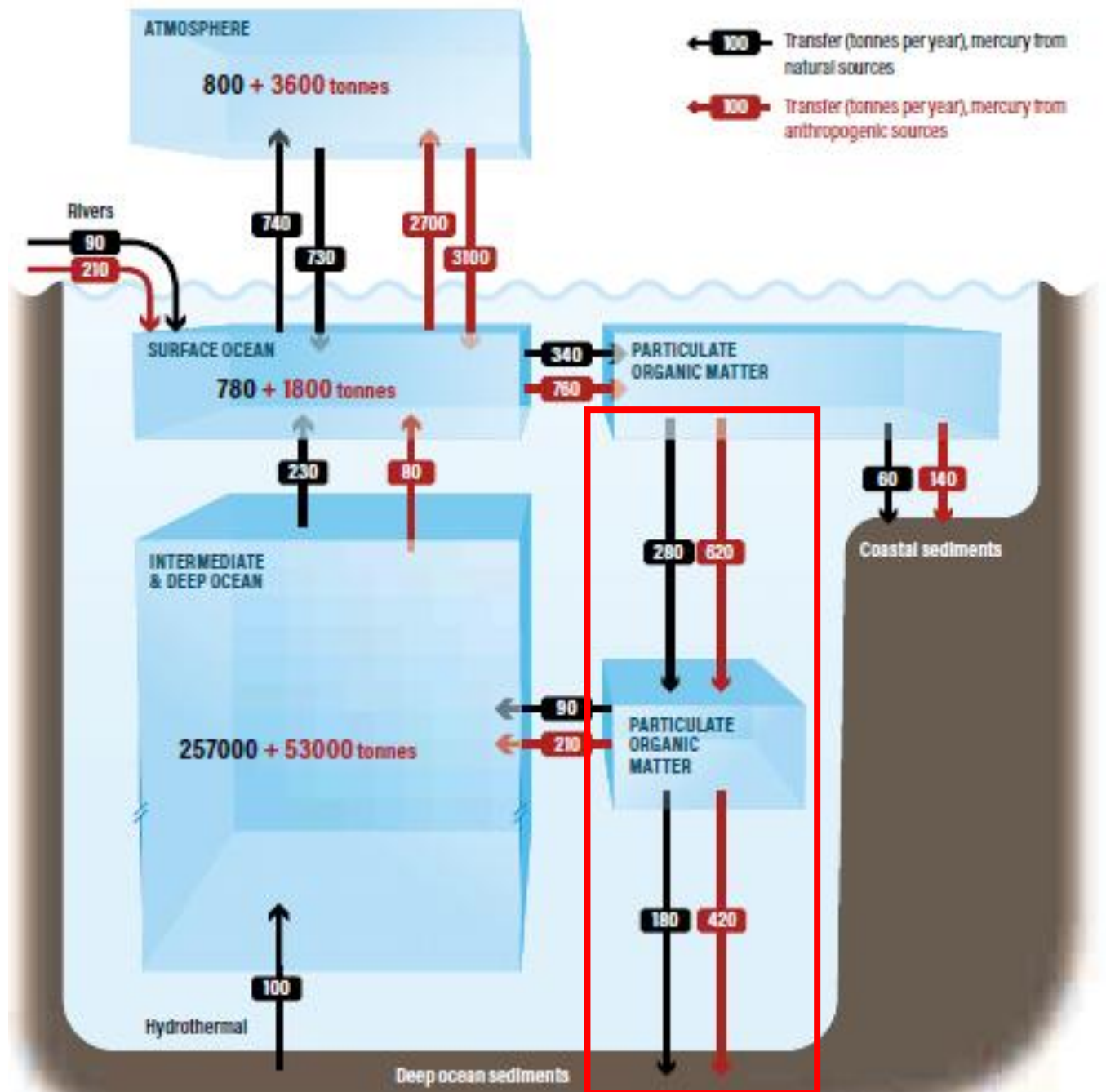
Reservoirs/burden: 10^8g Hg ; Fluxes: 10^8g Hg/yr ; Hg_p : mercury in particles



Current

Hg (in tonnes)





The amount of mercury removed by burial in deep ocean sediments is small compared to the amounts currently deposited at the surface of the ocean - **leads to a build up of mercury in ocean waters.**

Summary of the anthropogenic contributions to Hg cycle

- Human activities have increased total atmospheric mercury concentrations by **about 450% above natural levels.**
- **Combustion of fossil fuels and biomass is responsible for about 24% of the estimated global emissions**, primarily from coal burning (21%), besides other sources non-ferrous metal production (15%), cement production (11%), ferrous metal production (2%) and waste (about 7%) + gold mining industry
- Natural production of methylmercury in the oceans and in some lakes **is no longer limited by the input of inorganic mercury.**
- Mercury loads in some aquatic foodwebs are **at levels of concern for ecological and human health.**